ELITE

IGCSE Mathematics A • Pearson Edexcel

13_January_2023 • Unit 1

1H + 2H source-order past paper

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Contents

1H Q02 • Solving Linear Equations • 3 marks

1H Q07 • Probability Toolkit • 4 marks

1H Q08 • Circles, Arcs & Sectors • 5 marks

1H Q12 • Algebraic Roots & Indices • 2 marks

1H Q13 • Expanding Brackets • 3 marks

1H Q14 • Graphs of Functions • 3 marks

1H Q15 • Probability Diagrams - Venn & Tree Diagrams • 7 marks

1H Q16 • Sine, Cosine Rule & Area of Triangles • 5 marks

1H Q19 • Rounding, Estimation & Bounds • 5 marks

1H Q21 • Circles, Arcs & Sectors • 5 marks

1H Q23 • Coordinate Geometry • 6 marks

2H Q01 • Fractions • 3 marks

2H Q02 • Standard & Compound Units • 3 marks

2H Q03 • Set Notation & Venn Diagrams • 3 marks

2H Q04 • Area & Perimeter • 4 marks

2H Q05 • Rounding, Estimation & Bounds • 4 marks

2H Q06 • Coordinate Geometry • 3 marks

2H Q09 • Linear Graphs (y = mx + c) • 7 marks

2H Q10 • Right-Angled Triangles - Pythagoras & Trigonometry • 5 marks

2H Q11 • Graphs of Functions • 3 marks

2H Q13 • Algebraic Fractions • 3 marks

2H Q16 • Fractions, Decimals & Percentages • 2 marks

2H Q18 • Histograms • 5 marks

2H Q23 • 3D Pythagoras & Trigonometry • 5 marks

2H Q26 • Algebraic Roots & Indices • 5 marks

2 Solve $3(2 - 4x) = 5 - 8x$
Show clear algebraic working.

$x =$ ....

(Total for Question 2 is 3 marks)

7 Some members of a library were asked to name the type of book that they each liked to read the best.

One of the members is chosen at random.
The table shows information about the probability of the type of book that this member answered.

Type of book	comedy	romance	mystery	thriller
Probability	0.24	0.40	$3x$	x

48 members answered comedy books.

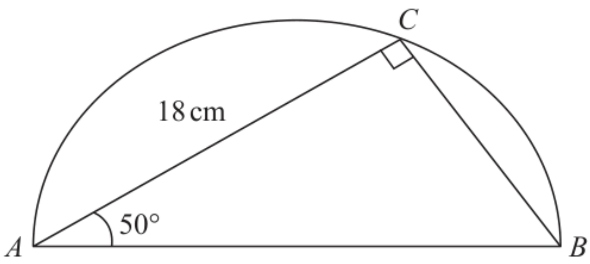
Work out how many of the members answered mystery books.

....

(Total for Question 7 is 4 marks)

8 The diagram shows a triangle *ABC* inside a semicircle.

Diagram **NOT** accurately drawn



A, *B* and *C* are points on the semicircle.

AB is the diameter of the semicircle.

Angle $ACB = 90°$
Angle $BAC = 50°$
$AC = 18$ cm

Work out the perimeter of the semicircle.
Give your answer correct to 2 significant figures.

.... cm

(Total for Question 8 is 5 marks)

12 $3^{\frac{1}{2}} \times 3^{\frac{2}{5}} = 3^m$

(a) Work out the value of m

$m =$ ....

(1)

$5^{-10} \div 5^{-4} = 5^n$

(b) Work out the value of n

$n =$ ....

(1)

(Total for Question 12 is 2 marks)

13 Expand and simplify $3x(2x - 5)^2$
Show clear algebraic working.

....

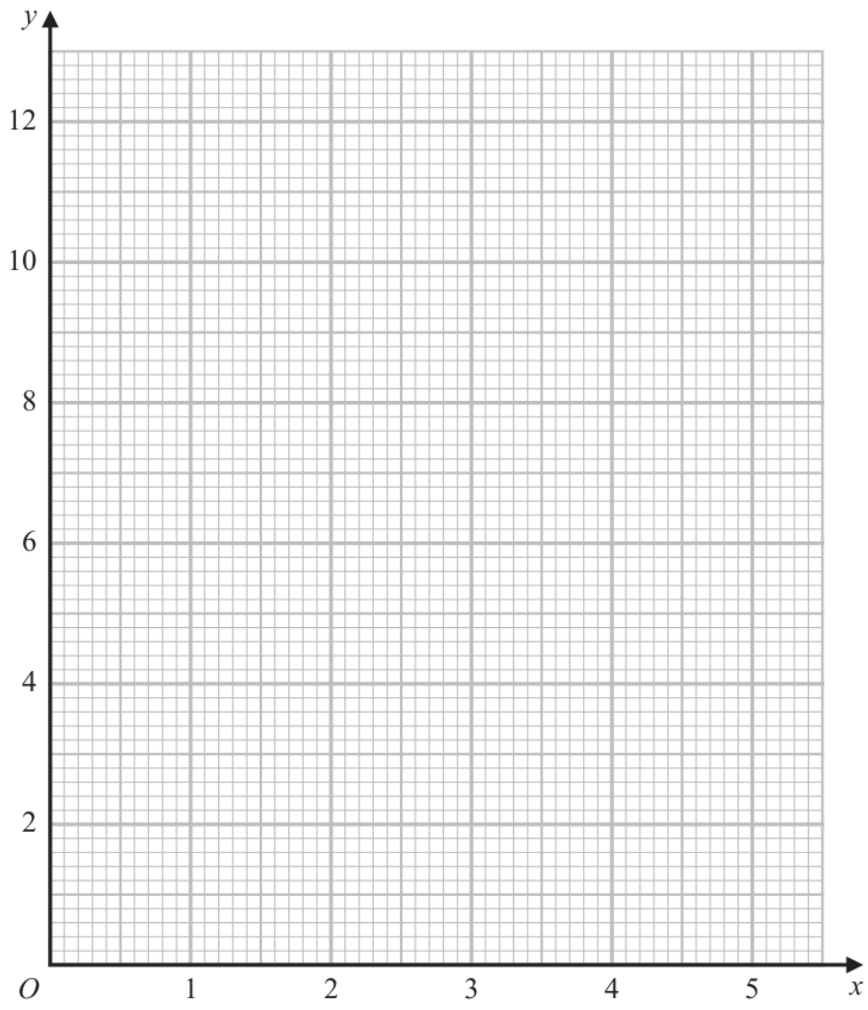
(Total for Question 13 is 3 marks)

14 (a) Complete the table of values for $y = \dfrac{2}{x}\left(5 - \dfrac{1}{x}\right)$

x	0.5	1	2	3	4	5
y		8		3.1	2.4	1.9

(1)

(b) On the grid, draw the graph of $y = \dfrac{2}{x}\left(5 - \dfrac{1}{x}\right)$ for $0.5 \leqslant x \leqslant 5$



(2)

(Total for Question 14 is 3 marks)

15 Here are 9 cards. Each card has either a number on it or a letter on it.

A	7	H	F	E	D	G	5	C

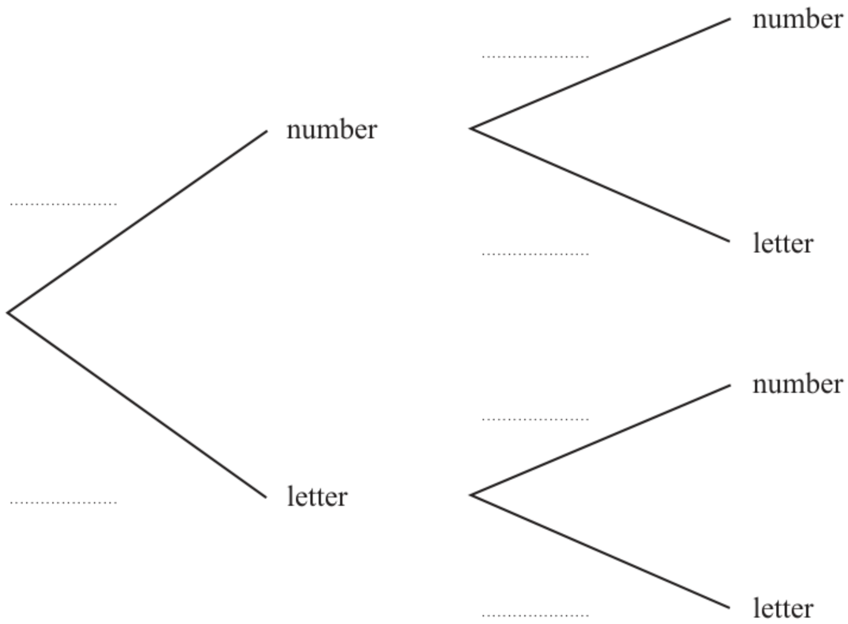
Tomas is playing a game.

Tomas takes at random one of the cards and keeps it.

Tomas then takes at random another card and keeps it.

(a) Complete the probability tree diagram.

First card **Second card**



(2)

(b) Work out the probability that each of the two cards has a number on it.

....

(2)

(c) Work out the probability that there will be one card with a number on it and one card with a letter on it.

....

(3)

(Total for Question 15 is 7 marks)

16 Here is a shape formed from two triangles *ABC* and *CDE*
ACD and *BCE* are straight lines.

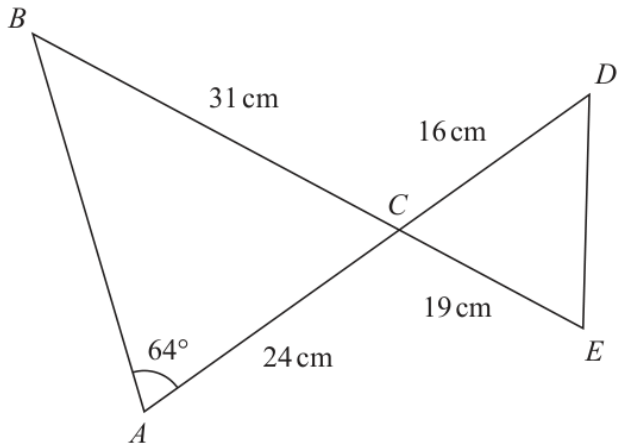


Diagram **NOT** accurately drawn

$AC = 24$ cm $\quad BC = 31$ cm $\quad CE = 19$ cm $\quad CD = 16$ cm

Angle $BAC = 64°$

Work out the length of *DE*
Give your answer correct to 3 significant figures.

.... cm

(Total for Question 16 is 5 marks)

19 The acceleration, a, of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places
$u = 5.12$ correct to 2 decimal places
$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a =$ ....

(Total for Question 19 is 5 marks)

21 The diagram shows the cross section of a circular water pipe.

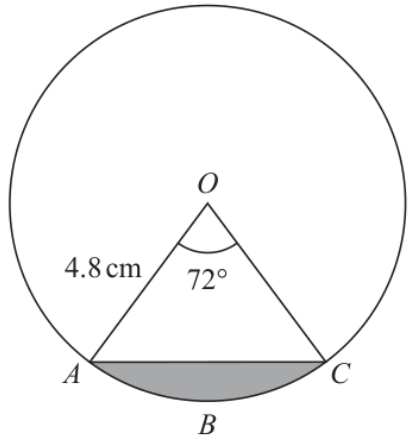


Diagram **NOT** accurately drawn

OABC is a sector of the circle, centre *O*

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.

.... cm^3

(Total for Question 21 is 5 marks)

23 *ABCD* is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D

The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

....

(Total for Question 23 is 6 marks)

1 Show that $3\frac{5}{7} \div 1\frac{5}{8} = 2\frac{2}{7}$

(Total for Question 1 is 3 marks)

2 Change a speed of 90 kilometres per hour to a speed in metres per second.
Show your working clearly.

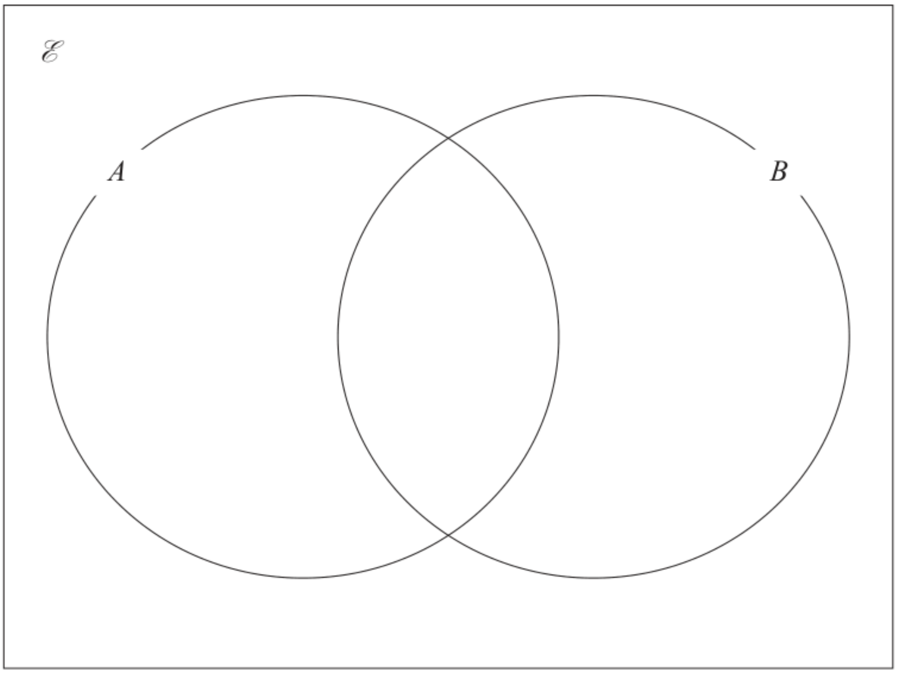
.... m/s

(Total for Question 2 is 3 marks)

3 $\mathscr{E} = \{11, 12, 13, 14, 15, 16, 17, 18, 19, 20\}$

$A = \{\text{even numbers}\}$
$A \cap B = \{12, 16, 20\}$
$(A \cup B)' = \{17, 19\}$

Complete the Venn diagram for the sets $\mathscr{E}$, A and B



(Total for Question 3 is 3 marks)

4 The diagram shows rectangle *ABCD*

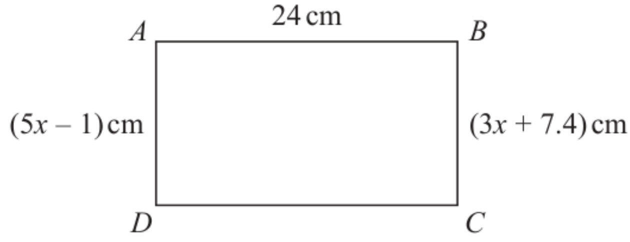


Diagram **NOT** accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

.... cm

(Total for Question 4 is 4 marks)

5 The weight of a cake is 2.75 kg, correct to 2 decimal places.

(a) Write down the lower bound of the weight of the cake.

.... kg

(1)

(b) Write down the upper bound of the weight of the cake.

.... kg

(1)

Penny has worked out $\dfrac{81.3 \times 59.2}{1.9^2}$ on her calculator.

Her answer is 13 332.299 17

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

(2)

(Total for Question 5 is 4 marks)

6 The points A and B are on a coordinate grid.

The coordinates of A are (6, 4)
The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j =$ ....

$k =$ ....

(Total for Question 6 is 3 marks)

9 (a) Write down the value of $(m + 2)^0$ where m is a positive integer.

....

(1)

(b) Simplify $(3a^2b^4)^3$

....

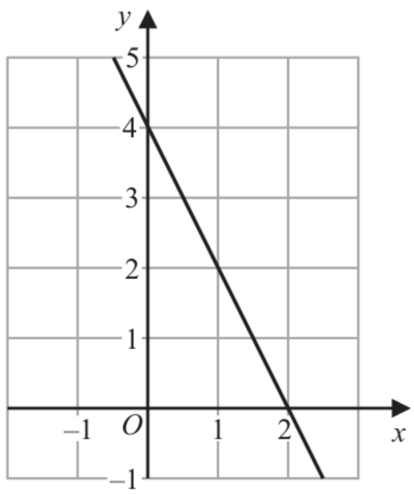
(2)

(c) Factorise fully $14x^2y^4 + 21x^3y^2$

....

(2)

The diagram shows a straight line drawn on a grid.



(d) Write down an equation of the line.

....

(2)

(Total for Question 9 is 7 marks)

10 The diagram shows an isosceles triangle, with base length 24 cm.

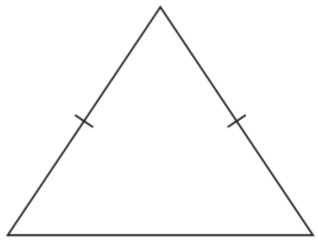


Diagram **NOT** accurately drawn

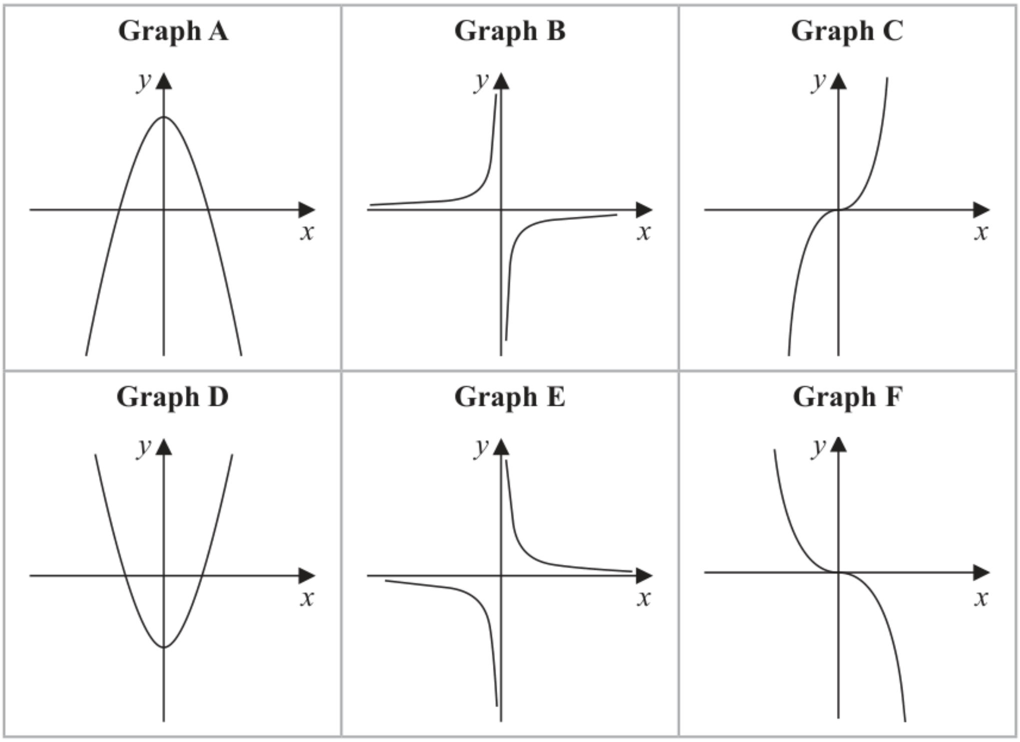
The perimeter of the triangle is 54 cm.

Work out the area of the triangle.

.... cm^2

(Total for Question 10 is 5 marks)

11 Here are six graphs.



Complete the table below with the letter of the graph that could represent each given equation.
Write your answers on the dotted lines.

Equation	Graph
$y = -\dfrac{2}{x}$	....
$y = 5 - x^2$	....
$y = -2x^3$	....

(Total for Question 11 is 3 marks)

13 Solve $\dfrac{x+3}{4} - \dfrac{7-x}{5} = 4.3$

Show clear algebraic working.

$x =$ ....

(Total for Question 13 is 3 marks)

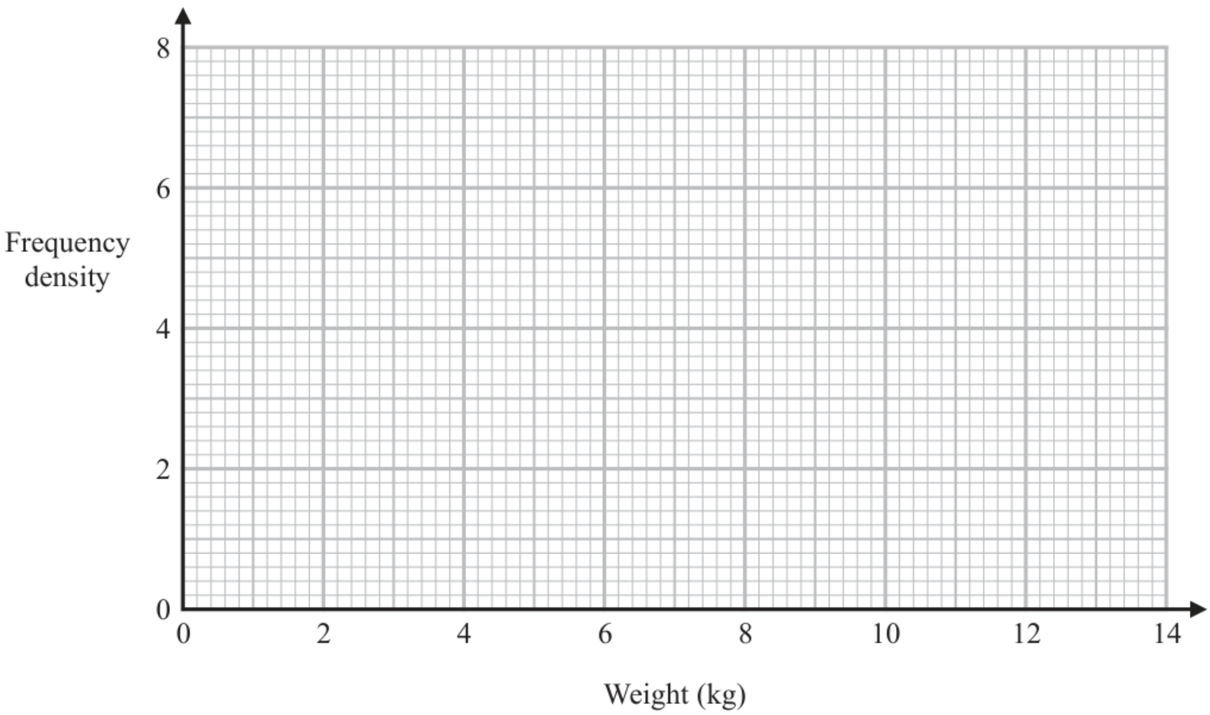
16 Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

(Total for Question 16 is 2 marks)

18 The table gives information about the weights, in kg, of the parcels that Pedro delivers on Monday.

Weight (w kg)	**Frequency**
$0 < w \leqslant 2$	12
$2 < w \leqslant 3$	7
$3 < w \leqslant 6$	15
$6 < w \leqslant 9$	12
$9 < w \leqslant 14$	9

(a) On the grid, draw a histogram for this information.



(3)

One of the parcels that Pedro delivered on Monday is chosen at random.

(b) Using the information in the table, find an estimate for the probability that this parcel weighs more than 7 kg.

....

(2)

(Total for Question 18 is 5 marks)

23 The diagram shows a solid prism *ABCDEFGHIJ*

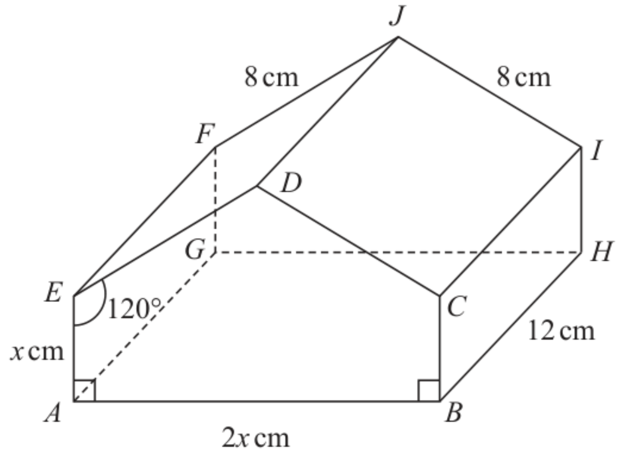


Diagram **NOT** accurately drawn

The prism is such that each cross section is a pentagon where

$AE = BC = x$ cm $\qquad AB = 2x$ cm $\qquad ED = CD = 8$ cm

angle EAB = angle $CBA = 90°$ $\qquad$ angle AED = angle $BCD = 120°$

Given that $AG = BH = EF = DJ = CI = 12$ cm

calculate the angle that AJ makes with the base $ABHG$ of the prism.
Give your answer correct to 3 significant figures.

°

(Total for Question 23 is 5 marks)

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

....

(Total for Question 26 is 5 marks)